

Imagine your system has a firewall, strong passwords, and everything looks secure.. but an attacker still gets inside using a valid login. That's exactly why modern security is shifting to a new model “Zero Trust”

What is Zero Trust Model

- Security model where no user/system is trusted by default
- Every access request must be verified
- Applies to both internal & external users

“Never Trust, Always Verify”

- Do not trust based on location
(inside network $\neq$ safe)
- Verify identity, device, and context every time
- Continuous authentication & authorization

Traditional vs Zero Trust

Traditional Security:

- Trust inside network
- Perimeter-based (firewall focused)
- Once inside → full access

Zero Trust:

- Trust nothing by default
- Identity-based security
- Continuous verification

Core Principles

Verify Explicitly

- Authenticate every request
- Use MFA, device checks

Least Privilege Access

- Give minimum required access
- Reduce attack surface

Assume Breach

- Design systems assuming attacker is already inside
- Monitor & limit damage